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A comparative analysis of large language models for the detection and classification of hate speech in a low-resource language

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ABSTRACT Hate speech represents one of the most significant challenges of modern digital society, particularly due to the pervasive use of social networks and online media. Developing reliable automated detection systems requires high-quality, meticulously curated, and annotated datasets - a task that is especially challenging for low-resource languages, such as Serbian. This paper describes the process of collecting, processing, and labeling textual data aimed at creating a dataset for hate speech detection in the Serbian language, as well as a comparative analysis of Large Language Models (LLMs) in the detection and classification of hate speech. The data were gathered from diverse sources, including social networks and online media platforms, utilizing both automated and manual techniques, as well as web crawling and scraping methods. The resulting dataset comprises 1351 short texts, containing 8029 sentences, annotated into three distinct classes: non-hate speech, offensive speech, and hate speech. Furthermore, hate speech instances were additionally categorized according to relevant types of discrimination in accordance with European Union legal acts. In addition to describing the annotation process, this paper provides a detailed analysis of the dataset, including class distribution, text length, and the most frequent keywords. Subsequently, the research selected five LLMs, which were inferred using prompt engineering with a different design for each approach. The models were evaluated using a literature-informed zero-shot prompting strategy based on detailed category definitions, decision criteria, and structured output constraints, implemented through single-stage, two-stage, and paragraph-level prompting settings. The selected models represent recent and widely used open-source general-purpose LLMs available through the Ollama platform, which was chosen to enable local, reproducible, and privacy-preserving evaluation on consumer hardware. Additional comparative experiments were also conducted using few-shot prompting, a general-purpose LLM, and the supervised BERTić model as external baselines. These LLMs were then employed for the detection and classification of hate speech, utilizing an ensemble strategy. The presented results highlight the complexity and diversity of hate speech in Serbian online communication, demonstrating a high detection accuracy of 86.7% achieved with the LLaMA 3 model, followed by Qwen (82%) for the two-stage sentence-level pipeline. For paragraph-level analysis, Qwen achieved the highest detection accuracy of 80%, slightly better than LLaMA 3 (77.3%). Ensemble of all five LLMs achieved comparable results to the best selected models, achieving 81.4% in two-stage sentence-level and 79.1% in paragraph-level analysis.

INDEX TERMS Artificial Intelligence, Hate Speech Detection, Large Language Models, Low-Resource Languages, Natural Language Processing

I. INTRODUCTION

Hate speech constitutes a form of public expression that incites, justifies, or spreads hatred, discrimination, and violence against individuals or groups based on their personal or social characteristics, in accordance with the definition provided by the United Nations [1], [2], [3]. These attributes may include national or ethnic origin, religion, race, gender, sexual orientation, disability, language, or any other actual or perceived characteristic. Unlike offensive or indecent speech, hate speech exerts a broader societal impact by directly threatening human dignity and potentially leading to manifest forms of discrimination and violence.

In contemporary society, the concept of hate speech has gained particular prominence due to the expansion of digital media and social networks. The Internet has enabled the exceptionally rapid dissemination of information without clear editorial barriers, creating an environment for the mass proliferation of various forms of harmful discourse. Anonymity and a perceived sense of impunity further embolden individuals to express views that they would otherwise refrain from sharing in traditional media or public spaces.

Today, hate speech is pervasive in online comments, social networks, forums, and video-sharing platforms. It is notably visible within the contexts of political debates, migration, inter-ethnic relations, particularly concerning sensitive regions such as Kosovo, a self-declared and UN-unrecognized country, and social crises, where it is often utilized as a tool for public opinion manipulation. While media outlets play a significant role in informing the public, their sensationalist approach or inadequate comment moderation sometimes contributes to the normalization of such discourse.

Identifying hate speech in a digital environment is a complex task, as the meaning of a message often depends on context, linguistic nuances, and cultural specificities. The same expression may be neutral in one context while serving a clear discriminatory or incitive function in another. Irony, sarcasm, and implicit messages introduce additional layers of complexity, making them difficult to recognize without a deep understanding of the content.

Traditionally, hate speech was identified manually by human moderators. Although this approach allows for high precision, it is time-consuming, costly, and unsustainable given the vast volume of content published daily on the Internet. Consequently, timely intervention in response to all controversial content is often impossible in practice. With the advancement of Artificial Intelligence (AI) and Natural Language Processing (NLP), automated systems for hate speech detection are increasingly being implemented [4], [5]. These systems utilize machine learning (ML) and deep learning (DL) techniques to analyze textual data and identify potentially problematic content, offering the advantage of speed and the capability to process large datasets in real-time [6].

However, automated hate speech detection faces significant limitations. Algorithms often struggle to distinguish between hate speech and freedom of expression, criticism, or satire, which can lead to false positives or negatives. A particular challenge arises with low-resource languages, where there is a lack of sufficient labeled data for high-quality model training. Therefore, modern systems increasingly employ a hybrid approach that combines automated analysis with human expertise or ensemble methods. In the domain of hate speech detection and classification, transformer architectures are commonly utilized. Some of the most prominent models trained for the Serbian language include bcms-bertic [7] (based on the ELECTRA architecture), as well as jerteh-81 [8], jerteh-355 [9], and SRoBERTa [10], based on the RoBERTa architecture. Beyond transformer architectures, Large Language Models (LLMs) can also be employed, even those not initially trained on Serbian or other low-resource natural languages.

The five evaluated LLMs were selected as recent and widely used open-source general-purpose models that were, at the time of the study, supported by the Ollama platform. This choice was made deliberately to ensure that all experiments could be executed locally under identical conditions, without reliance on external APIs, thereby supporting reproducibility, privacy, and deployment on consumer-grade hardware. The prompting strategy adopted in this study follows a literature-informed zero-shot design. Rather than relying on task-specific labeled demonstrations, the prompts guide the models through explicit hate speech category descriptions, decision criteria, and constrained output formats. This strategy was implemented through three complementary settings: single-stage sentence-level prompting, two-stage sentence-level prompting, and paragraph-level prompting. In this study, we provide a novel comparative benchmark of LLMs for Serbian hate speech detection. Unlike recent LLM-based hate speech studies that primarily benchmark prompting strategies across existing multilingual or high-resource datasets, this study focuses on Serbian as a low-resource language and combines local privacy-preserving inference, a newly compiled Serbian dataset, fine-grained category and subcategory classification, external supervised and general-purpose baselines, and statistical validation of selected model differences [11], [12], [13]. Our main contributions are summarized as follows:

- Hate speech detection benchmark: We establish a rigorous benchmarking framework for automated hate speech detection in the Serbian language, evaluating a diverse pipeline of local open-source LLMs. Through comprehensive comparative analysis against existing approaches, we demonstrate that our optimized models achieve state-of-the-art or highly competitive performance, showcasing the viability of locally deployed models for robust binary content moderation in low-resource settings.
- Fine-grained classification and subclassification: Moving beyond binary detection, we introduce a novel, highly granular taxonomy consisting of 14 categories and subcategories specifically tailored to capture the sociopolitical, cultural, and contextual nuances of hate speech in Serbian. While existing literature predominantly focuses on simple detection, this study provides one of the first deep multi-class classification frameworks for a low-resource Slavic language, addressing a critical gap in regional NLP research where such detailed subcategorization has not been previously explored.

- Methodological extension and prompt resilience: We extend the evaluation benchmark by introducing advanced comparative setups, integrating zero-shot and few-shot prompting techniques, a general-purpose commercial LLM baseline, and a fine-tuned, supervised transformer model (BERTić). This comprehensive framework demonstrates that our robust zero-shot baseline can be seamlessly extended into a few-shot learning paradigm, confirming the efficacy of our prompt design and proving that localized, prompt-based LLM inference offers a powerful, flexible, and scalable alternative to traditional classification paradigms.

The rest of the paper is organized as follows. Section II reviews related work on hate speech detection and classification, with a particular focus on approaches applied to low-resource languages and the use of LLMs. Section III describes the experimental setup, including dataset formation, annotation methodology, and the configuration of the evaluated LLMs. Section IV presents the proposed methodology, detailing the prompting strategies, sentence-level and paragraph-level analysis pipelines, and the ensemble-based inference framework. Section V reports and discusses the experimental results across different task formulations and model configurations. Finally, Section VI concludes the paper and outlines directions for future work.

II. RELATED WORK

Our research analyzes the detection and classification of hate speech in the Serbian language. As previously noted, Serbian is a low-resource language belonging to the South Slavic group, alongside Croatian, Slovenian, and Bosnian. Like other related languages, Serbian lacks a small or large language model (LLM), however, researchers have developed fine-tuned LLM such as YugoGPT [14]. Based on the Mistral-7B architecture with fine-tuning, YugoGPT was trained on a massive dataset comprising regional web content, literature, and media to better capture local context, grammar, and cultural nuances. It serves as a viable alternative to models like ChatGPT, which, despite supporting Serbian, often "think" in English and merely machine-translate output, leading to a loss of specific linguistic subtleties. While YugoGPT is better at recognizing specific insults and profanities, it lacks precision in robust hate speech detection and classification, as it is primarily superior in tasks requiring reasoning and response generation.

In contrast, BERTić was trained on a large, carefully curated corpus of 8 billion tokens specific to Serbia, Croatia, Montenegro, and Bosnia and Herzegovina, proving significantly superior to general multilingual models such as Multilingual Bidirectional Encoder Representations from Transformers (mBERT) [15]. While YugoGPT is a Decoder model designed for text generation and prompt engineering, BERTić functions as an Encoder model intended for classification, named entity recognition, and semantic analysis. In research, these two models are most frequently compared in the domains of text classification and contextual understanding [7]. BERTić has been extensively utilized for hate speech analysis, specifically through a specialized version called bcms-bertic-frenk-hate, based on the ELECTRA architecture. This model was fine-tuned on the FRENK dataset, which contains social media comments in Serbian and Croatian focusing on hate speech directed at migrants and the LGBTQ+ community [16]. Results indicate that BERTić is currently among the most precise models for this task in Serbian. Given its strong performance in Serbian and BCMS (Bosnian-Croatian-Montenegrin-Serbian) text classification, BERTić is treated in this study as a representative supervised state-of-the-art baseline for comparison with prompt-based LLM approaches. Other notable models for hate speech detection include Jerteh-81 and Jerteh-355, which are based on the RoBERTa architecture [8], [9].

Broader transformer-based and cross-lingual studies also show that transfer learning remains an important strategy for hate speech detection in low-resource or underrepresented languages. Recent work on cross-lingual hate speech detection demonstrates that transferring knowledge from pretrained language models can improve performance when target-language data are limited [17], while comprehensive transformer-era reviews emphasize persistent challenges related to dataset scarcity, domain shift, bias, and standardized evaluation [18]. These findings motivate the inclusion of prompt-based LLMs in the present comparison.

The critical need for automated detection tools to replace inefficient manual monitoring of Serbian online content is emphasized in recent research focusing on the safety and well-being of digital platform users. Cincović et al. [19] propose a comprehensive methodology for data collection and processing, highlighting that a robust detection model requires significant, diverse, and meticulously labeled datasets tailored to the linguistic specificities of the Serbian language. Furthermore, the study provides a conceptual framework for hate speech classification, serving as a foundational basis for developing scalable systems capable of mitigating real-world harm in the regional digital ecosystem. The hybrid model implemented in the described system is based on traditional ML algorithms and NLP techniques.

Earlier research on Serbian social media content utilized a combination of traditional ML algorithms, such as Support Vector Machines and Naive Bayes, to automate the classification of Twitter (later X) data [20]. While the study focused on sentiment and emotion detection, it established critical preprocessing pipelines for the Serbian language, addressing challenges like stemming and stop-word removal in a morphologically rich environment. These foundational efforts demonstrated the feasibility of automated text categorization in the regional digital space, providing a methodological basis for more advanced hate speech detection tasks.

In recent studies addressing the scarcity of resources for the Serbian language, researchers evaluated various ML and DL techniques using the AbCoSER dataset, a corpus consisting of annotated Serbian tweets [21]. Their findings indicate that the DL transformer architecture outperformed traditional algorithms, achieving a significant F1 macro score of 0.827 in identifying abusive language. This performance is considered highly competitive and aligns with established benchmarks for offensive speech detection in other languages with similar dataset magnitudes. Central to this achievement was the development of AbCoSER, the first manually annotated corpus for abusive speech in Serbian, which comprises 6,436 tweets with a specific sub-classification for hate speech instances. The researchers established a foundational abusive speech lexicon, enriched with linguistic triggers extracted directly from the dataset, providing a vital resource for enhancing the precision of automated detection and classification systems in the Serbian language [22].

Additional recent studies further show that reported hate speech detection performance varies strongly across datasets, modalities, and language settings. In a 2026 multimodal Indonesian study, XLM-RoBERTa achieved an F1-score of 0.803, multimodal LLMs such as Gemini 2.5 Flash and GPT-5.2 reached up to F1 = 0.799, and the best textualization-based multimodal setup achieved F1 = 0.833 [23]. In a 2026 low-resource Azerbaijani study, XLM-R achieved accuracy = 0.91 and macro-F1 = 0.72, while the corresponding Turkish high-resource setting reached accuracy = 0.94 and macro-F1 = 0.93, highlighting the effect of data scarcity [24]. In a recent LLM-based hate speech study with geographical contextualization, CodeLlama-7B achieved accuracy = 35.30%, and F1 = 52.18% [25].

Additional low-resource studies in other languages also report strong but highly dataset-dependent results. For Bengali, recent work comparing LLMs and transformer baselines reports that GPT-3.5 Turbo achieved an F1-score of 85.38%, while Gemini 1.5 Pro reached 86.17% for hate speech detection [26]. For Amharic, a recent DL study reported an F1-score of 94.8% using an SBi-LSTM model on a four-class hate speech dataset [27]. For Arabic, optimized BERT-based models have reported 93.2% accuracy / 82.0% F1 on OSACT-5, 85.0% accuracy / 87.0% F1 on LAHS, and 94.5% accuracy / 91.8% F1 on arHate [28]. These results provide additional reference points from other low-resource settings, although they are not directly comparable to the present study because of differences in language, dataset design, annotation scheme, and task formulation.

Recent work has also examined LLM-based hate speech detection more directly. Guo et al. [12] investigated several prompting strategies for context-aware hate speech detection across five established datasets, showing that prompt design strongly affects LLM performance. Ghorbanpour et al. [11] evaluated instruction-tuned multilingual LLMs across eight non-English languages and compared zero-shot and few-shot prompting with fine-tuned encoder models, reporting that prompt-based LLMs do not always outperform supervised encoders on real-world test sets. Similarly, Barakat et al. [13] systematically evaluated commonly used LLMs across multiple hate speech datasets, emphasizing that LLM-based moderation should be assessed across datasets and model families rather than through isolated single-model experiments. These studies demonstrate the relevance of LLM benchmarking for hate speech detection, but they primarily focus on established datasets, multilingual comparison, or binary detection settings, whereas the present study addresses Serbian-specific detection and fine-grained classification under local inference constraints.

Prompt engineering has also been explored for low-resource hate speech detection. Recent Bengali work compared zero-shot, multi-shot, role-based, and metaphor-based prompting with DL baselines [29]. The present study extends this direction by evaluating locally deployed open-source LLMs against BERTić and Gemini, while addressing not only binary detection but also category and subcategory assignment.

The scope of hate speech research in Serbian has also been extended to specialized domains, such as sports-related discourse on social media, where athletes are identified as a particularly vulnerable group. Recent experiments utilized a Bidirectional Long Short-Term Memory (BiLSTM) deep neural network for binary classification, achieving a high detection precision of up to 97% [30]. To overcome the lack of existing infrastructure, the authors developed a dedicated digital lexicon of hate speech terms and phrases specifically tailored to the sports domain in the Serbian language. While the model demonstrated exceptional precision, the results also highlighted the challenges of low recall, emphasizing the ongoing need for more diverse datasets and advanced architectures in domain-specific detection.

Beyond social and sports-related contexts, hate speech has emerged as a distinct subgenre of political discourse, frequently utilized in election campaigns in both Serbia and the USA to target vulnerable groups through vulgarisms and sexist rhetoric [31]. Research indicates that this type of language often functions as a tool for political manipulation, where information is strategically shaped using propaganda techniques to influence public behavior. By employing ML models to classify content from prominent Serbian media portals, such as RTS (public media service / national TV provider), Politika (the largest and oldest paper media in Serbia), and N1 (currently the only independent media in Serbia), studies have demonstrated the necessity of automated systems for identifying manipulative and hateful narratives in real-time [32]. These findings highlight how hate speech serves broader autocratic or partisan goals, further complicating the digital landscape through systematic ideological framing. A significant driver of the high prevalence of hate speech within Serbian discourse is its presence in political rhetoric, stemming from the Republic's leadership and government-aligned officials, as well as, to some extent, opposition politicians. This saturation poses a substantial challenge for dataset curation, as it leads to a disproportionate dominance of the political intolerance category, often overshadowing other forms of discriminatory speech. Consequently, achieving a balanced dataset requires meticulous filtering to ensure that the model can effectively generalize across all targeted groups without being biased toward political narratives.

In their comprehensive survey of multilingual hate speech detection, Marovac at el. analyze the evolution of models from cross-lingual embeddings to massively multilingual Transformers, emphasizing the persistent disparity between high-resource and low-resource languages [33]. The study explicitly identifies Serbian as a language that suffers from a significant lack of labeled datasets and specialized benchmarks, which hinders the development of robust localized detection systems. By categorizing existing research efforts, the paper highlights that while zero-shot transfer from English models offers a temporary solution, it often fails to capture the unique sociopolitical and cultural nuances inherent to the Serbian linguistic landscape. Consequently, the authors call for the creation of more diverse, human-annotated datasets to improve the performance of LLMs in underrepresented languages.

Research focusing on the Croatian language utilizes the NooJ algorithm to categorize slurs and insults while maintaining a critical ethical distinction between general offensive language and hate speech to prevent unjustified censorship [34]. The study highlights that informal online communication characterized by substandard syntax, spelling errors, and colloquialisms, poses a significant challenge for automated systems, leading to difficulties in minimizing false negatives. This linguistic complexity is particularly relevant given that Serbian and Croatian are highly similar and functioned as part of the same Serbo-Croatian language during the Yugoslav period, meaning that detection challenges and datasets often overlap across these regional contexts.

Research in the Serbian linguistic domain has expanded into the intersection of information theory and emotional speech analysis, using state-of-the-art ML models to estimate word-level surprisal values [35]. These studies demonstrate that language models can effectively predict speech parameters across different emotional states, even in low-resource environments. Such findings suggest that the computational representation of emotions and information density is viable for the Serbian language, providing a technical precedent for using similar architectures to identify the high-intensity emotional markers often found in hate speech.

Accelerating advances in NLP and big data analytics show that LLMs can effectively serve as automated annotators for short user-generated texts in the Serbian language, achieving results comparable to expert human labeling [36]. Comparative studies involving models such as GPT-5 and Gemini 2.5 Pro indicate that sophisticated fine-tuning allows these architectures to act as reliable alternatives to manual annotation in low-resource linguistic environments. Beyond simple labeling, specialized open-source systems now integrate web scrapers to extract and analyze formal news content from URLs, addressing the subtle and implicit hate speech often found in editorially framed articles [37]. These tools utilize prompt-based binary classifiers, such as LLaMA 3, to highlight hateful segments with high precision while maintaining the original text structure. Innovative frameworks have also been developed for content purification, where LLMs rewrite discriminatory speech into culturally appropriate and neutral reformulations [38]. By leveraging local execution environments like Ollama [39], these systems ensure data privacy while successfully preserving the original intent of the message without the harmful elements. This progress suggests that the strategic selection of LLMs and meticulous prompt design can significantly automate moderation and sanitization tasks for languages with limited digital resources.

The current lack of comprehensive resources for the Serbian language remains a major obstacle, as existing datasets fail to cover the broad spectrum of hate speech categories required for nuanced analysis. Most studies have overlooked the dual challenge of simultaneous detection and fine-grained classification, leaving the field without a unified analytical framework. At present, no highly accurate software tools exist for the Serbian market that utilize LLMs to automate the identification of such content in real-time. This research identifies the most effective LLM for this task, evaluates five different LLMs and an ensemble-based approach, and establishes a detection and classification framework for the Serbian language that can be readily adapted to other low-resource linguistic environments.

III. EXPERIMENTAL SETUP

All experiments were performed on a workstation equipped with a 13th Gen Intel Core i5-13400 processor (2.50 GHz), 32 GB of RAM, and an NVIDIA GeForce RTX 3060 GPU, running on an NVMe SSD under a standard Windows operating system environment. This configuration was used for running all models and ensuring consistent performance measurements during evaluation.

A. DATASET FORMATION AND ANNOTATION PROCESS

This section outlines the data collection process, the processing steps taken to create the annotated dataset, and the analysis of both the complete dataset and the partitioned sets for training, validation, and testing. Figure 1 illustrates this workflow.

Prior to collecting hate speech texts in Serbian, a foundational dataset was compiled in the form of an online dictionary containing offensive, profane, and derogatory words. This online dictionary was compiled from several already existing printed (hard copy) dictionaries, which were subsequently digitized using Optical Character Recognition (OCR) techniques. Utilizing this lexicon, the authors developed a Google Chrome web extension designed to censor such inappropriate language [40].

To facilitate the prevention of online hate speech through the training of AI models, careful consideration must be given to the training dataset. It is essential that the dataset be both sufficiently large and diverse to cover a broad spectrum of hate speech characteristics. The data collected for this study originate from various sources, including social media platforms - Meta’s Facebook and Instagram, X (formerly Twitter), YouTube, and Reddit. In addition to these, data were also gathered from daily news media platforms, namely Blic [41], Informer [42], N1 [43], and Sportal [44].

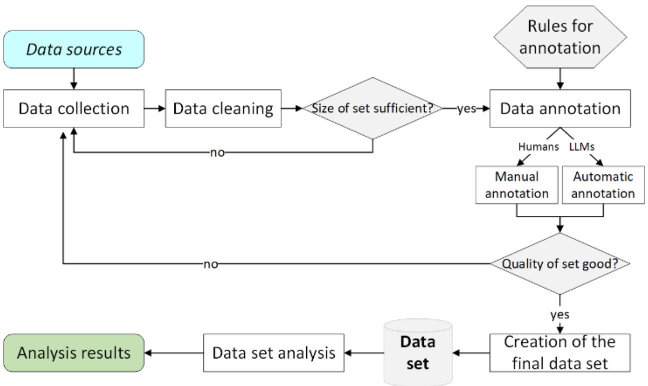

FIGURE 1. **Presentation of the data collection process with analysis.**

These sources were purposefully selected based on the requirements of this research, with the primary criterion being the probability of encountering hate speech. The aforementioned social networks have a high user base in Serbia, a wide range of content types, and active comment sections. Conversely, news platforms primarily share content within the domains of politics, economics, society, and sports - topics that frequently elicit offensive commentary. Where necessary, specific content on these platforms likely to contain hate speech was targeted to ensure a more balanced dataset.

Data collection was conducted through both automated and manual means. Automated data retrieval was facilitated via Application Programming Interfaces (APIs) where available, such as for Facebook, YouTube, and previously X. For daily news from media sites (Blic, Informer, N1, Sportal), specialized scripts were developed [37], specifically web crawlers and web scrapers, written in the Python programming language.

A small portion of the data was generated synthetically using LLMs, with the existing dataset serving as a reference. More specifically, the models were prompted to generate short texts containing hate speech. To improve control over generation, the prompts were extended with explicit definitions of hate speech and direct requests to include hate speech examples in the output. The models were also instructed to vary the length of the generated texts to increase diversity. However, even with various prompting techniques, LLMs were largely unable to generate high-quality data. Only a minimal amount of generated data (51 short texts were incorporated into final dataset) proved usable, as the outputs tended to be formulaic and repetitive. Furthermore, some online AI-powered applications and chatbots, such as ChatGPT [45], restrict the generation of hate speech in newer versions, while older versions often fail to produce satisfactory results for general use cases [46].

The final component of the data was sourced from the ReLDI-NormTagNER-sr 3.0 dataset, created by the CLARIN Knowledge Centre for South Slavic languages (CLASSLA) [47]. While this dataset was not originally intended for text classification, it contains an extensive collection of comments retrieved from X platform. Following the merging of these sources, the dataset underwent a cleaning process to remove comments that did not constitute meaningful linguistic units. The resulting dataset comprises 8029 sentences within 1351 short texts in the Serbian language. The distribution of samples across sources is shown in Figure 2. News media platforms constitute the largest portion of the dataset, with 973 samples (72.0%), followed by X platform with 153 samples (11.3%) and Reddit with 106 samples (7.8%). A smaller portion of the dataset was obtained through LLM-assisted synthetic generation, contributing 51 samples (3.8%), while Meta platforms (Facebook and Instagram) account for 41 samples (3.0%) and YouTube for 27 samples (2.0%).

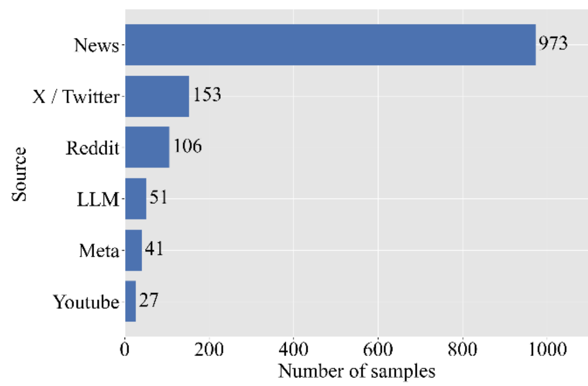

FIGURE 2. **Distribution of short texts across data sources.**

For experimental evaluation, the dataset was split at the paragraph level. The first 101 paragraph samples were used exclusively for testing, while the remaining samples were reserved for development purposes, including supervised training, validation, and prompt example selection in the additional comparative setups. Thus, all reported results were computed on a held-out test subset.

For the purposes of annotation, a system similar to the one presented by Davidson et al. [48] was adopted, wherein data are labeled with one of three primary classes: Class 0 – Neutral (Non-hate speech); Class U – Offensive speech; and Class 1-7 – Hate speech. The types of discrimination classified as hate speech include discrimination based on national, ethnic, religious, gender/sexual, political, and sports-fan affiliation, as well as any other form of dehumanization and degradation based on origin, occupation, physical appearance, abilities, or disability. The comprehensive list of categories and subcategories covered in this research is provided in Table 1, aligned with the European Union (EU) Code of Conduct on countering illegal hate speech online [49]. Furthermore, Table 2 provides illustrative examples of texts without hate speech, texts containing only insults, as a separate category, and texts with manifest elements of hate speech in the Serbian language.

It is important to note that a single sentence or a short multi-sentence text may express multiple categories of hate speech simultaneously. In contrast, offensive speech encompasses expressions such as profanity, vulgarities, and other individual insults, as well as a harsh or aggressive tone that may be unpleasant but is not inherently discriminatory. Thus, while all hate speech is inherently offensive, not all offensive speech constitutes hate speech.

TABLE I
PROPOSED CLASSIFICATION OF HATE SPEECH ACCORDING TO EU ACTS

Type of Hate Speech	Hate Speech Category and Subcategory Name
1: Racial, Ethnic, and National Hate Speech	1a: Race / Skin Color → Racism, Colorism 1b: Ethnic Affiliation → Ethnic discrimination, Ethnic hatred 1c: Nationality / Origin (Migration-based, Regional) → Xenophobia
2: Religious Hate Speech	Religion and Beliefs → Religious intolerance, religious discrimination
3: Gender and Sexuality-based Hate Speech	3a: Gender → Sexism 3b: LGBTQ+ Identity → Homophobia, Transphobia, Queerphobia
4: Physical Appearance and Health-based Hate Speech	4a: Physical Appearance → Lookism (discrimination based on physical appearance) 4b: Illness / Disability → Ableism
5: Age-based Hate Speech	Age (Youth, Elderly) → Ageism
6: Socio-economic Status-based Hate Speech	6a: Socioeconomic Status / Class → Classism 6b: Occupation / Profession → Stigmatization of professions 6c: Political intolerance
7: Sports-related Intolerance	Sports Affiliation (Fan Identity) → Sports-related intolerance, Hooliganism

Main categories are identified by the number preceding the category name, while subcategories, when multiple exist, are identified by lettered suffixes.

The following examples illustrate these distinctions:

- **Non-hate speech:** „Живео и здрав био у љубави и радости Никола Јокић" (in English: *"Long may he live in love and joy, Nikola Jokić"*);
- **Offensive speech:** „Зна ли се име тог потпуковника? Дајте објавите нека му потомци знају који је олош био" (in English: *"Is the name of that lieutenant colonel known? Publish it so his descendants know what scum he was"*);

Hate speech: „А што пуно причаш са тим лажовима? Поубијај ту четничку ђубрад" (in English: *"Why do you talk so much to those liars? Just kill those Chetnik scumbags"*).

A. DATASET DESCRIPTION

The dataset used in this study consists of paragraph-level textual samples, where each sample represents a coherent block of discourse composed of multiple sentences. Paragraphs were selected to reflect realistic language usage patterns commonly encountered in online discussions, comment sections, and informal public communication, which frequently contain a mixture of neutral, offensive, and hate-related content. This design choice allows the dataset to capture both isolated and context-dependent instances of hate speech within a single unified corpus.

TABLE II
EXAMPLES OF CATEGORIES AND SUBCATEGORIES FOR THE SERBIAN HATE SPEECH DETECTOR

Input Data (Serbian / *English*)	Output Data	
	Class	Explanation
Sviđa mi se što su Kinezi radilice i što će završiti pre roka. / *I like that the Chinese are hard workers and that they will finish ahead of schedule.*	0	neutral sentence
Is that your sister? You both look like retards. / *Is that your sister? You both look like retards.*	U	offense
Ne želimo Cigane u našoj zemlji, treba ih ukloniti. / *We don't want Gypsies in our country, they should be removed.*	1a	racism
Mi nismo divlji kao Šiptari, niti velikoustaše kao Hrvati. / *We are not savage like Albanians, nor extreme Ustashas like Croats.*	1b	ethnic discrimination
Ja neću zapošljavati divlje Albance sa Kosova. / *I will not employ wild Albanians from Kosovo.*	1c	xenophobia
Sve adventističke grupe su lažovi koji nas jure po ulici. / *All Adventist groups are liars who chase us down the street.*	2	religion and beliefs
Žene su preslabe za liderske uloge. / *Women are too weak for leadership roles.*	3a	sexism
Gej populacija je odvratna. / *The gay population is disgusting.*	3b	homophobia
Ružni ljudi ne zaslužuju pažnju. / *Ugly people don't deserve attention.*	4a	lookism
Osobe sa invaliditetom su beskorisne. / *Disabled people are useless.*	4b	ableism
Tinejdžeri su glupi i neodgovorni. / *Teenagers are stupid and irresponsible.*	5	ageism
Ta bedna rulja bi samo da deli tuđe pare, a prstom ne bi mrdnuli. / *That miserable mob only wants to hand out other people's money, and they wouldn't lift a finger themselves.*	6a	classism
Novinari su lažovi. / *Journalists are liars.*	6b	profession
Ekipa iz vlasti nema hrabrost da osudi ustastvo i to je istina. / *The people in power do not have the courage to condemn Ustashism, and that is the truth.*	6c	political intolerance
Svi navijači Partizana su kreteni. / *All Partizan fans are jerks.*	7	sports-related intolerance

English translation is given for better understanding, in italic font.

The structural properties of the dataset are illustrated through the distribution of sentence counts per paragraph, shown in Figure 3. Most paragraphs are relatively short, with the highest concentration observed between approximately three and six sentences. As paragraph length increases, the number of samples gradually decreases, producing a right-skewed distribution with a pronounced long tail. While the majority of samples remain below ten sentences, the dataset also contains a smaller number of substantially longer paragraphs, extending beyond twenty and in rare cases above thirty sentences. This variability enables the evaluation of model performance across both compact inputs and more context-rich, extended discourse units.

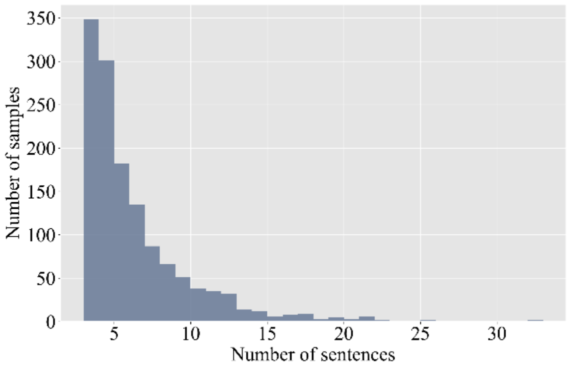

FIGURE 3. Distribution of sentences per paragraph in the dataset.

Hate speech is not uniformly distributed across sentences within paragraphs. Figure 4 presents the distribution of the number of hate-labeled sentences per paragraph. While a substantial portion of paragraphs contains no hate speech (27.7%, 374 out of 1351), the remaining 72.3% (977 out of 1351) samples most often include one to three hate-labeled sentences embedded within otherwise neutral or offensive discourse. Paragraphs containing larger numbers of hate speech sentences become progressively less frequent, but are deliberately included to represent sustained or escalating hostile narratives. This distribution supports the evaluation of sentence-level discrimination within a broader textual context.

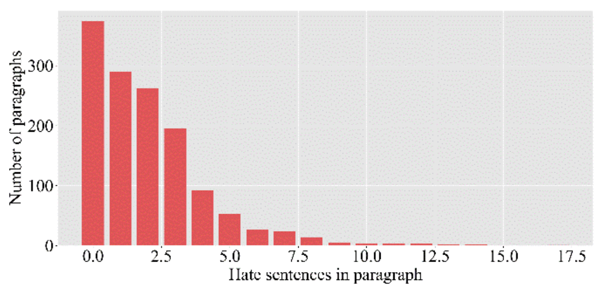

FIGURE 4. Distribution of hate labeled sentences per paragraph in the dataset.

At the sentence level, the dataset exhibits a three-way class distribution between non-hate content, offensive language, and hate speech, as shown in Figure 5. The majority of sentences are labeled as non-hate, while the remainder is divided between offensive content and explicit hate speech. Offensive language is treated as a distinct category to differentiate general insults or profanity from speech targeting protected groups. This separation enables more precise analysis of model behavior in borderline cases.

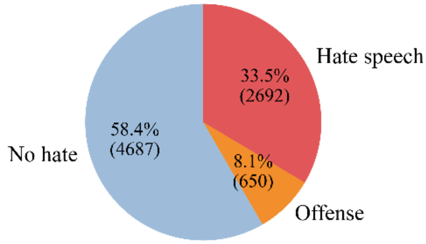

FIGURE 5. Sentence-level class distribution.

Fine-grained category analysis is derived from sentence-level annotations. Figure 6 presents the distribution of hate speech sentences across the main category taxonomy. Racial, Ethnic, and National Hate Speech constitutes the largest group, followed by Socio-economic Status-based Hate Speech and Gender and Sexuality-based Hate Speech, while Sports-related Intolerance is also substantially represented. Physical Appearance and Health-based Hate Speech, Religious Hate Speech, and Age-based Hate Speech occur less frequently, but still provide coverage of multiple hate speech types for comparative evaluation.

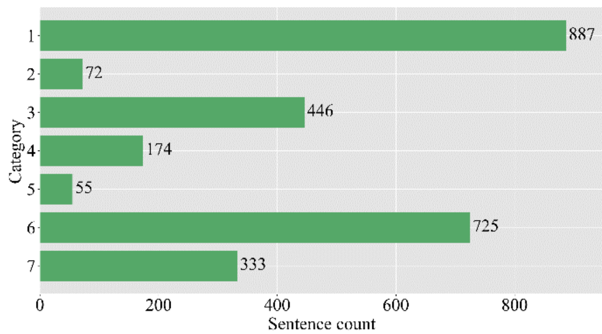

FIGURE 6. Distribution of hate speech sentences across the main category taxonomy.

A more detailed breakdown by subcategory is shown in Figure 7. Ethnic discrimination and xenophobia dominate the subcategory distribution, followed by political intolerance and sexism. Other prominently represented subcategories include stigmatization of professions, socioeconomic status / class discrimination, homophobia, transphobia, and queerphobia, and lookism, while ableism and racism / colorism occur less frequently. This distribution reflects the diversity of hate speech phenomena covered in the dataset and further motivates the use of LLMs capable of generalizing across both frequent and less frequent subcategories.

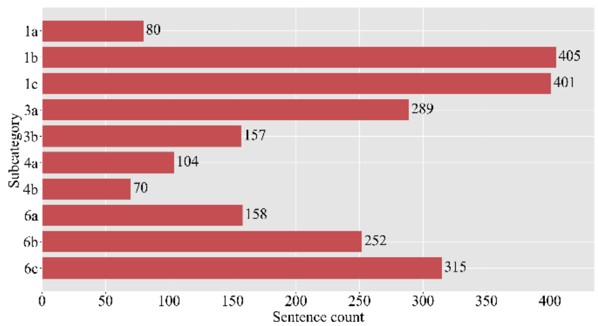

FIGURE 7. **Distribution of hate speech sentences across subcategories.**

The annotation process involved three annotators. Two primary annotators (A1 and A2) independently labeled all 1,351 paragraphs at the sentence level, assigning with no hate speech category code (number 0), hate speech category codes (1-7 with subcategories) or marking offensive but non-hate-speech content. Both annotators were graduate students in computational linguistics with prior experience in hate speech annotation for the Serbian language. Table III summarizes the interrater agreement between A1 and A2 across 8,029 sentence-level annotations, measured using Cohen's Kappa [50] and Krippendorff's Alpha [51], [52]. All agreement scores exceed 0.95, indicating almost perfect interrater agreement according to the standard interpretation of Cohen's Kappa. At the sentence level, only 248 out of 8,029 sentences (3.1%) received differing labels. A senior annotator (A3), a researcher with over five years of experience in NLP and content moderation, served as a tiebreaker, providing annotations only for the 132 samples (9.8%) where A1 and A2 disagreed.

TABLE III
INTERRATER AGREEMENT BETWEEN ANNOTATOR 1 AND ANNOTATOR 2 ACROSS 8,029 SENTENCE-LEVEL ANNOTATIONS AT FOUR GRANULARITY LEVELS

Granularity	Cohen's κ	Krippendorff's α
Binary (hate vs. no hate)	0.963	0.963
Three-way (hate / offense / no hate)	0.963	0.963
Top-level category (0-7)	0.962	0.962
Exact code match	0.953	0.953

Table IV presents an illustrative example of a paragraph-level sample from the dataset together with its human annotation. The example demonstrates how a single paragraph may contain sentences exhibiting different linguistic phenomena, including offensive language, multiple hate speech categories, and neutral content. As described in Section III.A, annotations are performed at the sentence level, while the paragraph itself is treated as a single dataset sample.

In the presented example, the first sentence is labeled as offensive language without hate speech. The second sentence contains hate speech targeting both socioeconomic status (classism) and ethnic identity. The third sentence continues the same hostile narrative toward protected groups. The fourth sentence does not contain hate speech and is labeled as neutral. The final sentence expresses political intolerance, illustrating that political hate speech may co-occur with other categories within the same paragraph. This example highlights the motivation for paragraph-level modeling and sentence-level annotation, as well as the complexity of real-world discourse where heterogeneous labels may appear within a single textual unit.

TABLE IV
EXAMPLE OF A PARAGRAPH-LEVEL DATASET SAMPLE AND ITS HUMAN ANNOTATION

Language	Example of one sample in dataset	Human annotation
Serbian	Ovu odvratnu Dijanu Hrku bih pitala kako nije okacila ni jednu sliku sina na svojim drustvenim mrezama sve dok nije poginuo??? Ponaša se kao neka prosjakinja i ciganka, i smatra da sada svi treba da je sažaljevamo. A znamo da nije bila u dobrim odnosima sa sinom. I ovi drugosrbijanci što je podržavaju su sramota ljudskog roda.	
English (transl.)	*I would like to ask this disgusting Dijana Hrka why she did not post a single photo of her son on her social media until he died??? She behaves like some kind of beggar and gypsy, and thinks that everyone should now feel sorry for her. And we know that she was not on good terms with her son. And those "other-Serbians" who support her are a disgrace to the human race.*	U,(6a;1b),0,6c

The corresponding labels for this paragraph follow the taxonomy defined in Section III.A and reflect the presence of offensive language, socioeconomic and ethnic hate speech, neutral content, and political intolerance within a single sample. This example illustrates why paragraph-level inputs are essential for evaluating context-aware hate speech detection and classification.

B. LARGE LANGUAGE MODEL CONFIGURATIONS

This section presents the LLMs selected for evaluation. Five open-source LLMs were chosen based on their widespread adoption in the research community, accessibility through the Ollama platform, and ability to run on consumer hardware through quantization techniques. The key specifications of each model, including the number of parameters, context window length, and storage size, are summarized in Table V. These metrics provide insight into the computational requirements and capabilities of each model, enabling informed selection based on specific application needs and available hardware resources.

As shown in Table V, the selected models differ substantially in parameter count, context window, and quantization profile, which enables a later analysis of whether hate speech detection performance is driven primarily by prompt design or by the capabilities of the underlying model. In particular, differences in model scale and available context capacity may affect how reliably each model can perform fine-grained classification, while prompting strategies determine how effectively those capabilities are utilized for the target taxonomy. Among the evaluated models, significant variations exist across all measured specifications, reflecting the diverse design philosophies and intended use cases of each model.

TABLE V
MODEL COMPARISON

Model	Params (B)	Context	Size (GB)
Phi-3	3.8	128K	2.2
Qwen3	8.2	40K	5.2
DeepSeek-R1	8.2	128K	5.2
Llama 3	8.0	8K	4.7
Mistral	7.2	32K	4.4

Phi-3, developed by Microsoft [53], stands out as the most lightweight option with only 3.8 billion parameters and a storage requirement of 2.2 GB. Despite its compact size, Phi-3 offers the largest context window of 128K tokens, matching the capability of DeepSeek-R1. This combination of minimal resource requirements and extensive context handling makes Phi-3 particularly suitable for deployment in resource-constrained environments.

Qwen3, DeepSeek-R1, and Llama 3 share similar parameter counts in the 8 billion range, representing a common configuration in open-source LLMs that balances capability with accessibility. However, these models exhibit significant variation in their context length capabilities.

Llama 3, developed by Meta [54], provides the most limited context window at 8K tokens, which may constrain its applicability for tasks requiring extensive context, such as document summarization or multi-turn conversations.

Qwen3, developed by Alibaba [55], extends this capability to 40K tokens, providing five times the context capacity of Llama 3.

DeepSeek-R1, developed by DeepSeek [56], further extends this capability to 128K tokens, representing a 16-fold increase compared to Llama 3 and enabling processing of substantially longer documents and conversations.

Mistral, developed by Mistral AI [57], occupies a middle position among the evaluated models with 7.2 billion parameters and a 32K token context length. The 4.4 GB storage requirement positions Mistral as a moderately sized option that can be deployed on standard consumer hardware while providing competitive capabilities.

The storage sizes across the evaluated models do not scale linearly with parameter counts. This can be attributed to differences in quantization methods employed. Qwen3, DeepSeek-R1, and Mistral utilize Q4_K_M quantization, a mixed-precision approach that retains higher precision for certain critical layers. In contrast, Phi-3 and Llama 3 employ Q4_0 quantization, a uniform 4-bit scheme that achieves maximum compression at the potential cost of some performance degradation.

The context window length is a critical factor as it determines the scope of information a model can consider during inference. Models with larger context windows can maintain coherence across longer documents and engage in extended multi-turn dialogues without losing track of earlier context. However, larger context windows also increase computational requirements, as attention mechanisms scale quadratically with sequence length. Therefore, the selection of an appropriate context length should be guided by the specific requirements of the target application.

IV. METHODOLOGY

This section describes the methodological framework used to evaluate LLMs for hate speech detection and classification in low-resource languages. The proposed novel methodology is designed to systematically compare multiple prompting strategies, input granularities, and model configurations, while ensuring reproducibility and consistency across experiments. The implementation, datasets, and evaluation scripts used in this study are publicly available at https://github.com/mdodovic/HateSpeech-LLM-Detector

The proposed system performs hate speech analysis using instruction-prompted LLMs. Given an input text, the system determines whether hate speech is present and, if so, assigns one or more predefined categories and subcategories introduced in Section III. The methodology supports both binary hate speech detection and fine-grained multi-class classification, reflecting realistic moderation and annotation scenarios.

All experiments were conducted using locally hosted LLMs via the Ollama service [39], which enabled controlled execution, consistent inference conditions, and avoidance of external API variability. Each model was executed with identical prompts, decoding parameters, and input preprocessing to ensure comparability across results.

The system relies on instruction-driven inference using LLMs, where task behavior is fully specified through carefully designed prompts. Rather than fine-tuning models, the system relies on prompt engineering to guide the models toward consistent detection and categorization of hate speech, which is particularly suitable for low-resource language settings where annotated data is limited [58].

Each interaction with a model begins with a global system prompt that defines the model's role and constraints. This system prompt establishes the model as an assistant for hate speech detection and enforces concise and precise responses. In all experiments, the following system-level instruction was used without modification, presented in Table VI.

All task-specific prompts build upon this system instruction and differ only in task structure and input granularity. In prompts that require category assignment, the category definitions and subcategory explanations described in Sections III.A and III.B are explicitly injected at runtime through a placeholder mechanism denoted as `{categories_prompt}`. This placeholder is replaced with the full textual description of all hate speech categories and subcategories prior to inference, ensuring that each model receives identical semantic grounding for classification decisions. We adopted a zero-shot prompting strategy whose prompt design was literature-informed and grounded in established prompting methodology, emphasizing detailed category descriptions and decision criteria without including task-specific labeled examples [59][60]. This prompting methodology is named "New Tasks Without Extensive Training" [61].

TABLE VI
SYSTEM PROMPT

Language	Prompt content
Serbian	Vi ste pomoćnik za detekciju govora mržnje. Odgovarajte sažeto i tačno, prateći instrukcije.
English (transl.)	*You are an assistant for hate speech detection. Respond concisely and accurately, following the instructions.*

Three approaches were presented: two-stage sentence analysis, single-stage sentence analysis, and paragraph-level analysis. All three approaches were executed independently for every model, allowing direct comparison of model behavior across task granularities and prompt structures.

In the two-stage sentence-level setting, each sample consists of a single sentence treated as an independent unit. This setup reflects common annotation practices in hate speech datasets and allows controlled evaluation of model behavior on short, potentially ambiguous inputs. Two different prompting strategies were evaluated.

The first strategy decomposes the task into two sequential steps: detection followed by classification. The diagram representing this approach is presented in Figure 8. In the detection step, the model is asked to determine whether the input sentence contains hate speech according to a definition that explicitly references discriminatory or pejorative language directed toward individuals or groups based on protected attributes. The detection prompt used in this stage is shown in Table VII.

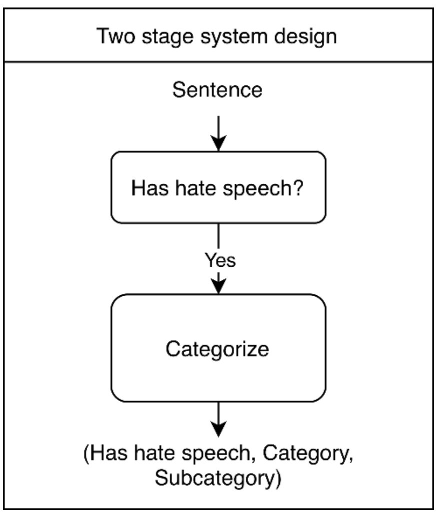

FIGURE 8. Two-stage sentence-level execution diagram.

TABLE VII
TWO-STAGE DETECTION PROMPT

Language	Prompt content	Output pattern
Serbian	Analiziraj sledeći tekst i odredi da li sadrži govor mržnje. Govor mržnje je jezik koji napada ili koristi pežorativne ili diskriminatorne izraze u odnosu na osobu ili grupu na osnovu rasnih, etničkih, polnih/rodnih, verskih, seksualnih, zdravstvenih, nacionalnih ili drugih svojstava. Tekst: {text}; Odgovori sa "DA" ili "NE" i dodaj kratko objašnjenje. Odgovor:	(has_hs,)
English (transl.)	*Analyze the following text and determine whether it contains hate speech. Hate speech is language that attacks or uses pejorative or discriminatory expressions toward an individual or a group based on racial, ethnic, gender/sexual identity, religious, sexual orientation, health-related, national, or other characteristics. Text: {text}; Respond with "YES" or "NO" and add a brief explanation. Answer:*	

If the model outputs a positive detection decision, the same sentence is passed to a second prompt that performs fine-grained categorization. This classification prompt constrains the model to return only category and subcategory codes, thereby preventing explanatory text and ensuring machine-readable outputs. The classification prompt is defined in Table VIII.

TABLE VIII
TWO-STAGE CLASSIFICATION PROMPT

Language	Prompt content	Output pattern
Serbian	{categories_prompt}; Analiziraj sledeći tekst i svrstaj ga u jednu ili više kategorija iznad. Odgovori isključivo nizom kodova (bez dodatnog teksta), gde je svaki kod broj kategorije 0–7 uz opcioni slovni sufiks podkategorije (npr. 4a, 6a, 2). Ako tekst pripada više kategorija, kodove obavezno odvoji zarezom bez razmaka. Tekst: {text}; Odgovor:	(class_id,)
English (transl.)	*{categories_prompt}: Analyze the following text and assign it to one or more of the categories listed above. Respond exclusively with a sequence of codes (without any additional text), where each code consists of a category number from 0–7 with an optional alphabetical subcategory suffix (e.g., 4a, 6a, 2). If the text belongs to multiple categories, the codes must be separated by commas without spaces. Text: {text}; Answer:*	

The detection prompt (Table VII) produces a single output variable, `has_hs`. This variable represents the model's binary decision on whether the input text contains hate speech. In implementation, `has_hs` is treated as a Boolean value derived from the model's "YES/NO" answer (Serbian

"DA/NE"). The accompanying short explanation required by the prompt is used only as a justification for the decision and is not used to derive additional labels. The classification prompt (Table VIII) outputs `class_id`. This variable encodes the predicted hate-speech label as a compact code corresponding to the taxonomy defined in Section III. The numeric part of `class_id` denotes the main category (0–7), while an optional alphabetical suffix denotes the subcategory (e.g., `1a`, where `1` is the category and `a` is the subcategory). If multiple categories apply, the model is instructed to return multiple `class_id` codes as a comma-separated list with no spaces (e.g., `4a,6a,2`).

This single-stage sentence-level formulation allows the detection and classification tasks to be evaluated independently and enables analysis of error propagation between stages. The diagram of this phase is presented in Figure 9.

This strategy performs detection and classification jointly using a single prompt. In this formulation, the model is required to output exactly two lines: one indicating the presence or absence of hate speech and one specifying the corresponding categories. The strict output format is intended to reduce ambiguity and enforce consistency across models. The combined prompt is presented in Table IX.

TABLE IX
SINGLE-STAGE DETECTION AND CLASSIFICATION PROMPT

Languag e	Prompt content	Output pattern
Serbian	{categories_prompt}; Analiziraj sledeći tekst. Odgovori ISKLJUČIVO taplom sa TAČNO DVA POLJA. Prvo polje GovorMržnje: DA ili NE; Drugo polje: Kategorije: lista kodova odvojenih zarezom BEZ razmaka. Ako je odgovor NE, napiši samo 0. Tekst: {text}; Odgovor:	(has_hs, class_id)
English (transl.)	*{categories_prompt}: Analyze the following text. Respond EXCLUSIVELY with EXACTLY TWO FIELDS. First field: HateSpeech: YES or NO Second field: Categories: a list of codes separated by commas WITHOUT spaces. If the answer is NO, write only 0. Text: {text}; Answer:*	

This approach evaluates whether joint reasoning within a single inference step improves classification reliability and reduces inconsistencies observed in multi-step pipelines.

The single-stage prompt (Table IX) produces a pair of outputs: `(has_hs, class_id)`. The first output, `has_hs`, is the same binary indicator as in Table VII, derived from the "YES/NO" (or "DA/NE") field. The second output, `class_id`, follows the same encoding rules as in Table VIII. When `has_hs` is negative (NO/NE), the prompt enforces a special case in which `class_id` must be set to `0`, indicating the absence of hate speech and therefore the absence of any applicable category.

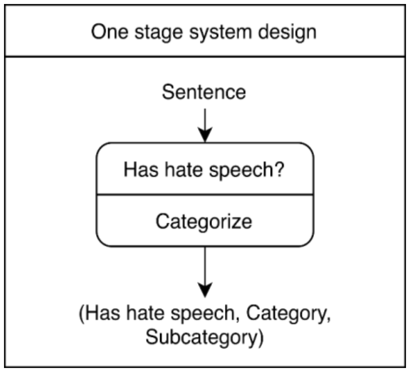

FIGURE 9. Single-stage sentence-level execution diagram.

In addition to sentence-level analysis, the methodology includes paragraph-level evaluation to assess the influence of broader textual context. In this setting, each input sample consists of a paragraph containing multiple sentences, where hate speech may be present in only a subset of them, as described in Section III.B. This strategy is presented in the diagram on the Figure 10.

The model is instructed to segment the paragraph into sentences without modifying their content and to assign a hate speech category to each sentence individually. This formulation allows the model to leverage inter-sentential context while preserving sentence-level labels. The prompt used for paragraph-level analysis is given in Table IX.

TABLE X
PARAGRAPH-LEVEL DETECTION AND CLASSIFICATION PROMPT

Languag e	Prompt content	Output pattern
Serbian	{categories_prompt}; Podeli dati tekst na rečenice. Nemoj menjati, preformulisati ili kombinovati rečenice. Za SVAKU rečenicu dodeli kategoriju (0-7). Odgovori tako što ćeš svaku rečenicu navesti u POSEBNOM REDU u formatu: (rečenica; kategorija; podkategorija); Tekst: {text}; Odgovor:	(sentence_id, has_hs, cat_id, subcat_id)
English (transl.)	*{categories_prompt}: Split the given text into sentences. Do not modify, rephrase, or combine sentences. Assign a category (0–7) to EACH sentence. Respond by listing each sentence on a SEPARATE LINE in the following format: (sentence; category; subcategory); Text: {text}; Answer:*	

This approach enables context-aware classification, particularly in cases where hate speech is implicit, referential, or dependent on surrounding discourse.

In addition to individual model evaluation, an ensemble-based inference mechanism was implemented to assess whether aggregating multiple model predictions improves robustness. For each input sample, all models were executed concurrently using multi-threaded inference. The final decision was derived using majority voting over hate speech detection outcomes and assigned categories.

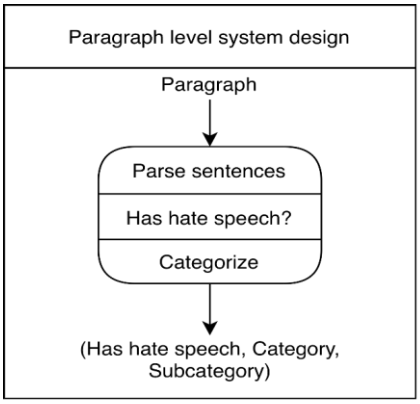

FIGURE 10. **Paragraph-level execution diagram.**

This ensemble strategy was designed to mitigate model-specific biases and variability, which are particularly pronounced in low-resource language scenarios. By combining heterogeneous model outputs, the methodology evaluates whether consensus-based decisions provide more stable and reliable hate speech classification compared to individual models.

The paragraph-level prompt (Table X) outputs a structured record per sentence, represented in the output pattern as `(sentence_id, has_hs, cat_id, subcat_id)`. Here, `sentence_id` identifies the sentence within the paragraph after sentence segmentation performed by the model. The variable `has_hs` again denotes whether that particular sentence contains hate speech. Unlike the sentence-level settings, where category and subcategory are merged into a single code, paragraph-level outputs are separated into `cat_id` and `subcat_id`. The `cat_id` is the integer category identifier (0-7), while `subcat_id` is the subcategory suffix (e.g., `a`, `b`) when applicable; if no subcategory applies or if `cat_id = 0`, `subcat_id` is left empty.

To complement the core experimental framework, the following subsection introduces additional comparative setups, including few-shot prompting, general-purpose LLM evaluation, and the BERTić baseline.

A. ADDITIONAL COMPARATIVE SETUPS

To complement the main prompt-based configurations described in the previous subsections, we additionally considered three comparative setups aimed at broadening the evaluation perspective. These setups were introduced to examine whether performance can be improved through example-guided prompting, to compare locally deployed open-source LLMs with general-purpose LLMs, and to benchmark the proposed prompt-based approach against a supervised encoder model specialized for the BCMS linguistic space. This design follows recent methodological recommendations in LLM-based hate speech detection, where prompt-based results are more informative when compared against supervised encoders, alternative prompting strategies, and strong general-purpose models rather than interpreted in isolation [11], [62], [63].

For the supervised baseline, we fine-tuned BERTić [7], which we treat as a representative state-of-the-art Serbian/BCMS encoder baseline, for three tasks: binary hate speech detection, top-level category classification, and fine-grained subcategory classification. To ensure direct comparability with the prompt-based experiments, all three setups used the same sentence-level prediction objective and the same dataset split protocol, as described in section III.A.

In all BERTić experiments, the model was trained using a sentence-in-context strategy [64], [65]. Each instance was formed as a pair consisting of the full paragraph and the target sentence, where the paragraph provided broader discourse context and the sentence represented the unit to be classified. This design was chosen because annotations were assigned at the sentence level, while the interpretation of hate speech often depends on the surrounding paragraph context.

To improve robustness under class imbalance, all BERTić variants were trained using weighted cross-entropy loss and regularized with dropout = 0.1, weight decay = 0.01, and label smoothing = 0.1, while model selection was based on validation F1-score with early stopping to reduce overfitting. A stratified validation split of 0.15 was used within the training portion of the dataset to preserve class distribution. After preliminary hyperparameter tuning, the final reported configuration was kept fixed across all three tasks and used 10 epochs, batch size 8, learning rate 5×10^{-5}, maximum sequence length 512, gradient accumulation over 4 steps, and warmup ratio 0.15, with no layer freezing applied in the final setup. As an additional robustness check for the supervised baseline, BERTić was also evaluated using stratified 5-fold cross-validation on the development portion of the dataset. The original held-out test subset was excluded from this procedure, preserving its role as the common test set for direct comparison with the prompt-based LLM experiments. Stratification was performed with respect to the target label of each task, and the same modeling strategy and hyperparameter configuration were used across folds. The cross-validation results are reported as mean ± standard deviation across the five folds.

For the general-purpose LLM comparison, we employed Gemini 3 Flash [66] and adopted the same prompting strategy as in Radenković et al. [36], keeping the task structure, category definitions, and output format aligned with the rest of the study. This ensured that the comparison remained focused on differences between locally deployed open-source LLMs and a strong general-purpose proprietary model, rather than on changes in prompt construction. Gemini 3 is part of Google's current Gemini model family and represents a suitable reference point for this additional comparison.

For the few-shot comparative setup, we considered only the two-stage sentence-level pipeline, since this configuration provided the clearest separation between binary detection and fine-grained categorization in the main experiments. This two-stage few-shot categorization workflow is illustrated in Figure 11. In this setting, the Stage 1 detection prompt remained unchanged and was evaluated in the same zero-shot form as in Table VII, while few-shot prompting was introduced only in Stage 2 (categorization step). More specifically, the categorization prompt followed the same structure as the prompt shown in Table VIII, but was extended with several labeled demonstration examples written in the format sentence; categories; explanation. The selected demonstrations were designed to cover representative decision patterns relevant to the task, including: sentences with no hate speech, insults classified as 0 (not hate speech), hate speech belonging to a single category, hate speech spanning multiple categories, and borderline cases in which a sentence may plausibly be interpreted both as containing hate speech and as not containing it. This design allowed the model to preserve the original zero-shot detection behavior while receiving additional guidance only for the more difficult category assignment stage, where fine-grained distinctions and output formatting are more challenging.

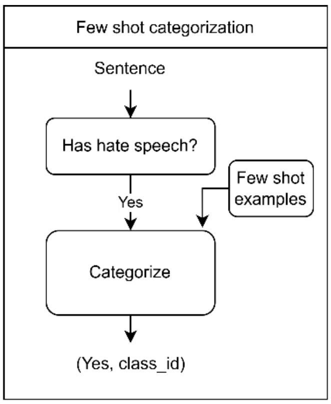

FIGURE 11. **Two-stage sentence-level few-shot categorization diagram.**

V. RESULTS AND DISCUSSION

This section presents and analyzes the experimental results obtained from sentence-level hate speech detection and classification using both single-stage and two-stage prompting strategies. The evaluation considers five LLMs: LLaMA, Mistral, DeepSeek, Phi, and Qwen, and compares their performance across binary detection, category-level classification, and subcategory-level classification, as well as inference latency. In all following tables presenting achieved metrics scores and measured inference time, the best value in each column is highlighted in bold. Specifically, higher values indicate better performance for Accuracy and F1-score, whereas lower values indicate better performance for the reported inference time (ms).

A. SINGLE-STAGE SENTENCE-LEVEL RESULTS

Table XI summarizes the sentence-level performance of the single-stage prompting approach, in which hate speech detection and fine-grained classification are jointly performed within a single model inference. The evaluation includes five individual LLMs as well as an ensemble configuration constructed from their outputs.

For binary hate speech detection, the single-stage approach achieves generally strong performance across models. Qwen attains the highest detection accuracy and F1 score, indicating robust discrimination between hate and non-hate sentences. LLaMA and DeepSeek also demonstrate competitive detection capability, while Mistral and Phi exhibit noticeably lower F1 values, reflecting reduced recall and overall stability. The ensemble further improves detection robustness, achieving high accuracy and balanced F1 performance, which confirms that combining model outputs mitigates individual model biases and variance.

In contrast, category-level classification performance is substantially weaker in the single-stage setting. Although Mistral achieves the highest category-level accuracy and F1, all models (including the ensemble) exhibit relatively low scores, indicating that jointly performing detection and high-level category assignment in a single stage remains challenging. This effect is most pronounced for Phi and DeepSeek, whose category-level performance drops sharply despite acceptable detection accuracy. Mistral achieves the highest category-level accuracy and F1, while the ensemble provides more stable mid-range performance than several individual models.

A similar trend is observed for subcategory-level classification, where performance further degrades across individual models. LLaMA and Qwen achieve the strongest performance among the single models, and the ensemble is comparable; nevertheless, overall scores remain modest, indicating that fine-grained subcategory discrimination is particularly challenging in the single-stage setting.

From a computational perspective, inference time varies considerably across models. Lightweight models such as Mistral and Phi achieve the fastest response times, while DeepSeek and Qwen incur substantially higher latency due to larger context processing and more complex reasoning. The ensemble approach, while yielding superior classification accuracy, introduces the highest total inference cost, as expected, making it less suitable for real-time deployment scenarios.

To sum up, the results indicate that while the single-stage approach is effective for binary hate speech detection, it struggles with reliable category and subcategory classification, particularly at finer granularity. Overall, the ensemble provides competitive performance and can offer more stable predictions than some individual models, suggesting that output aggregation may be beneficial in single-stage prompting. Nevertheless, the single-stage formulation, where detection and taxonomy assignment are tightly coupled, still yields modest category and subcategory scores, indicating limited reliability for nuanced hate speech taxonomy. This motivates the adoption of two-stage strategies, examined in the following subsection.

TABLE XI
SINGLE-STAGE SENTENCE-LEVEL HATE SPEECH DETECTION AND CLASSIFICATION RESULTS

Model	Detection		Classification of categories		Classification of subcategories		Time [ms]
	Accuracy	F1	Accuracy	F1	Accuracy	F1	Total
llama	0.705	0.727	0.429	0.429	**0.479**	**0.479**	467.39
mistral	0.607	0.471	**0.458**	**0.458**	0.143	0.143	**175.67**
deepseek	0.770	0.650	0.185	0.185	0.385	0.385	10138.27
phi	0.639	0.313	0.074	0.074	0.200	0.200	518.97
qwen	**0.836**	**0.792**	0.276	0.276	0.421	0.421	7640.80
ensemble*	0.803	0.700	0.385	0.385	0.429	0.429	14887.38

*: represents an ensemble of all models, not the name of a specific model

B. TWO-STAGE SENTENCE-LEVEL RESULTS

Table XII presents the performance of the evaluated LLMs under the two-stage sentence-level pipeline, where hate speech detection and category classification are executed as separate inference steps. Compared to the single-stage approach, this design aims to reduce task interference by isolating the binary decision from fine-grained classification.

In the detection stage, several models improve substantially compared to the single-stage configuration, most notably LLaMA and Mistral, while Qwen remains comparable and the ensemble shows mixed behavior. LLaMA attains the strongest detection performance among individual models, with an accuracy of 0.867 and an F1 score of 0.875, indicating robust discrimination between hate and non-hate content. Qwen follows with 0.820 accuracy and 0.784 F1, and DeepSeek achieves 0.784 accuracy and 0.777 F1. Phi remains a weak detector (low detection F1), while Mistral's detection F1 is comparable to other strong detectors, despite lower accuracy.

In terms of efficiency, the two-stage pipeline incurs significantly higher latency compared to the single-stage approach, primarily due to the additional classification pass. DeepSeek and Qwen exhibit the largest inference times, exceeding 18 seconds per sample in total, while LLaMA and Phi remain comparatively more efficient. Despite this overhead, the two-stage configuration demonstrates a clear trade-off favoring classification accuracy over inference speed.

TABLE XII
TWO-STAGE SENTENCE-LEVEL HATE SPEECH DETECTION AND CLASSIFICATION RESULTS

Model	Detection		Classification of categories		Classification of subcategories		Time [ms]		
	Accuracy	F1	Accuracy	F1	Accuracy	F1	detect	categorize	total
llama	**0.867**	**0.875**	0.333	0.333	0.429	0.429	**1459.37**	722.60	2181.97
mistral	0.733	0.778	0.300	0.300	0.261	0.261	1487.99	767.04	2255.03
deepseek	0.784	0.777	0.625	0.625	0.667	0.667	9267.36	10861.31	20128.66
phi	0.672	0.286	0.499	0.490	**0.750**	**0.750**	1574.81	**492.58**	**2067.39**
qwen	0.820	0.784	0.556	0.556	0.521	0.507	7284.95	10845.84	18130.79
ensemble*	0.814	0.600	**0.721**	**0.721**	0.623	0.623	11313.13	12642.73	23955.86

*: represents an ensemble of all models, not the name of a specific model

Overall, the results indicate that the two-stage architecture is particularly advantageous improving category-level and subcategory-level classification for most models, particularly DeepSeek, Qwen, and the ensemble, while detection performance remains strong for the best-performing detectors (notably LLaMA) but varies across models. These findings highlight the importance of task decomposition when applying LLMs to structured annotation problems in low-resource languages.

The ensemble approach further amplifies these gains. By aggregating predictions from all models via majority voting, the ensemble achieves the highest category-level classification performance (F1 0.721) and strong subcategory performance, though DeepSeek and Phi achieve higher subcategory F1. This confirms that model diversity contributes positively to fine-grained hate speech categorization, particularly in low-resource language settings where individual models may exhibit complementary strengths. However, this improvement comes at the cost of increased computational overhead, with the ensemble recording the highest total inference time. The separation of detection and classification yields a notable improvement in category-level classification across most models. DeepSeek exhibits a substantial gain, reaching a category-level F1 score of 0.625, while Qwen achieves comparable performance with an F1 score of 0.556. These results suggest that models benefit from being conditioned only on hate-positive samples during the classification stage, reducing confusion introduced by neutral or offensive-only content. Subcategory classification follows a similar trend, with DeepSeek achieving the highest F1 score among single models (0.667), indicating improved sensitivity to fine-grained distinctions once the detection task is resolved.

C. PARAGRAPH-LEVEL RESULTS

Table XIII reports the performance of LLMs on paragraph-level hate speech detection and classification. In this setting, each input consists of a multi-sentence paragraph containing a mixture of neutral, offensive, and hate speech content, requiring the models to exploit broader contextual information when producing predictions. Unlike sentence-level evaluation, paragraph-level analysis reflects more realistic discourse scenarios, where hate speech may be embedded within longer narratives.

For binary hate speech detection, overall performance remains comparable to sentence-level results, although a slight decrease in accuracy and F1 score is observed for several models. Qwen achieves the strongest detection accuracy at 0.800, indicating robust identification of hate-related content even when diluted across multiple sentences. LLaMA and the ensemble also demonstrate stable detection performance, while Phi and DeepSeek show reduced consistency, particularly in recall-sensitive cases. These results suggest that extended context can introduce ambiguity, especially for models with limited long-context reasoning capacity.

TABLE XIII
PARAGRAPH-LEVEL HATE SPEECH DETECTION AND CLASSIFICATION RESULTS

Model	Detection		Classification of categories		Classification of subcategories		Time [ms]
	Accuracy	F1	Accuracy	F1	Accuracy	F1	Total
llama	0.773	**0.667**	0.682	0.682	0.200	0.188	4718.80
mistral	0.727	0.429	0.727	0.727	0.254	0.254	4283.84
deepseek	0.656	0.462	0.545	0.568	0.205	0.205	16910.62
phi	0.516	0.571	0.467	0.467	0.400	0.400	**2447.50**
qwen	**0.800**	0.571	0.756	0.756	**0.667**	**0.667**	18078.88
ensemble*	0.791	**0.667**	**0.763**	**0.763**	0.333	0.333	32007.56

*: represents an ensemble of all models, not the name of a specific model

At the category-level classification, paragraph-level context leads to notable performance improvements compared to sentence-level classification. Qwen and the ensemble achieve the highest category-level accuracy and F1 scores, exceeding 0.75, demonstrating that access to surrounding sentences provides valuable contextual cues for identifying the dominant hate category. Mistral also performs competitively at this level, while DeepSeek and Phi show weaker results, indicating sensitivity to noise introduced by mixed-content paragraphs.

In contrast, subcategory-level classification remains challenging at the paragraph level. While Qwen again outperforms other models with an F1 score of 0.667, most models exhibit substantial degradation relative to category-level performance. The ensemble approach, although effective for category prediction, achieves only moderate subcategory accuracy, suggesting that majority voting may obscure fine-grained distinctions when subcategories vary across sentences within the same paragraph. This highlights a fundamental limitation of paragraph-level aggregation for detailed hate speech taxonomy.

From an efficiency standpoint, inference time increases significantly for paragraph-level inputs. Models such as DeepSeek and Qwen incur the highest computational costs due to longer input sequences and more complex reasoning requirements. While Phi and Mistral remain relatively efficient, their reduced classification accuracy limits their practical usefulness for paragraph-level moderation. The ensemble configuration, while achieving strong classification performance, exhibits the highest latency, making it more suitable for offline analysis rather than real-time applications.

Overall, the results in Table XII demonstrate that paragraph-level analysis enhances category-level hate speech classification by leveraging contextual information across sentences. However, this benefit does not fully extend to subcategory discrimination, where overlapping or conflicting signals within paragraphs introduce additional complexity.

To further contextualize the obtained results, the following subsection compares the proposed prompt-based settings with additional prompting strategies and external baselines.

D. COMPARATIVE ANALYSIS WITH ADDITIONAL BASELINES, PROMPTING STRATEGIES, AND STATISTICAL VALIDATION

Table XIV summarizes the results obtained with the additional comparative setups introduced in Section IV.A, namely the supervised BERTić baseline, the general-purpose LLM baseline, and the few-shot categorization variants of the two-stage sentence-level pipeline. For the few-shot setting, binary detection results remained unchanged, since the first stage was kept in the original zero-shot form and few-shot examples were introduced only in the categorization stage. Therefore, the binary metrics for Qwen3 and Llama 3 correspond to the same two-stage detection results reported earlier in the paper. To assess the reliability of reported performance differences given the limited test set, Table XIV also reports 95% bootstrap confidence intervals [67] for all configurations included in this additional comparison. These intervals were estimated using nonparametric bootstrap resampling with replacement over the held-out test samples, based on 10,000 resampling iterations, and are used to support a more cautious interpretation of differences between zero-shot, few-shot, supervised, and general-purpose model settings.

TABLE XIV
ADDITIONAL COMPARATIVE RESULTS FOR BERTIĆ, THE GENERAL-PURPOSE LLM BASELINE, AND TWO-STAGE FEW-SHOT CATEGORIZATION

Model	Detection		Classification of categories		Classification of subcategories	
	Accuracy†	F1†	Accuracy†	F1†	Accuracy†	F1†
General-purpose LLM	0.801 [0.792, 0.810]	0.632 [0.615, 0.649]	0.460 [0.410, 0.510]	0.460 [0.410, 0.510]	0.421 [0.402, 0.440]	0.421 [0.402, 0.440]
BERTić	0.783 [0.748, 0.818]	0.770 [0.727, 0.808]	0.207 [0.201, 0.214]	0.352 [0.316, 0.393]	0.192 [0.109, 0.266]	0.217 [0.205, 0.230]
Zero-shot qwen	0.820 [0.814, 0.827]	0.784 [0.777, 0.791]	0.556 [0.540, 0.575]	0.556 [0.540, 0.575]	0.521 [0.513, 0.530]	0.507 [0.496, 0.520]
Zero-shot llama	0.867 [0.856, 0.880]	0.875 [0.859, 0.894]	0.333 [0.292, 0.377]	0.333 [0.292, 0.377]	0.429 [0.417, 0.442]	0.429 [0.417, 0.442]
Few-shot qwen	0.820 [0.814, 0.827]	0.784 [0.777, 0.791]	0.909 [0.866, 0.947]	0.727 [0.692, 0.762]	0.817 [0.750, 0.889]	0.714 [0.678, 0.755]
Few-shot llama	0.867 [0.856, 0.880]	0.875 [0.859, 0.894]	0.571 [0.535, 0.612]	0.462 [0.431, 0.490]	0.385 [0.374, 0.397]	0.286 [0.251, 0.321]

†: Values in brackets represent 95% bootstrap confidence intervals estimated over 10,000 resampling iterations

To further assess whether selected differences between models were statistically meaningful, McNemar's [68] exact test was applied to paired correctness outcomes on the same held-out samples. The test was performed for selected binary, category-level, and subcategory-level comparisons, with statistical significance assessed at $p < 0.05$. The results are summarized in Table XV.

The McNemar's test results provide additional statistical context for the main comparative observations. At the binary level, statistically significant differences were observed between the local zero-shot LLMs and the external BERTić and general-purpose LLM baselines, while the difference between zero-shot LLaMA and zero-shot Qwen was not significant. At the category level, zero-shot LLaMA and zero-shot Qwen differed significantly, and the few-shot Qwen setting showed a statistically significant improvement over zero-shot Qwen. Few-shot Qwen also differed significantly from few-shot LLaMA at the category level. In contrast, none of the selected subcategory-level comparisons reached statistical significance.

TABLE XV
MCNEMAR'S EXACT TEST RESULTS FOR SELECTED PAIRWISE COMPARISONS

Comparison	Level	p-value	Sig. †
ZS LLaMA vs. ZS Qwen	binary	0.8592	No
ZS LLaMA vs. BERTić	binary	0.0016	Yes*
ZS LLaMA vs. GP-LLM	binary	0.0148	Yes*
ZS Qwen vs. BERTić	binary	0.0098	Yes*
ZS Qwen vs. GP-LLM	binary	0.0150	Yes*
ZS LLaMA vs. ZS Qwen	category	0.0290	Yes*
ZS LLaMA vs. ZS Qwen	subcategory	0.4731	No
FS LLaMA vs. ZS LLaMA	category	1.0000	No
FS Qwen vs. ZS Qwen	category	0.0215	Yes*
FS Qwen vs. FS LLaMA	category	0.0094	Yes**
FS LLaMA vs. ZS LLaMA	subcategory	0.0708	No
FS Qwen vs. ZS Qwen	subcategory	0.7201	No
FS Qwen vs. FS LLaMA	subcategory	0.7552	No

ZS = zero-shot; FS = few-shot; GP-LLM = general-purpose LLM.
†: Sig. = Significance. Star markers indicate $*p < 0.05$ and $**p < 0.01$, based on McNemar's exact test.

The additional comparison shows that BERTić remained competitive for binary hate speech detection, with an accuracy of 0.783 and an F1-score of 0.770, but performed considerably worse at finer granularity levels, reaching 0.207 accuracy and 0.352 F1 for category classification, and 0.192 accuracy and 0.217 F1 for subcategory classification. The general-purpose LLM baseline achieved somewhat better fine-grained results than BERTić, with 0.460 accuracy/F1 at the category level and 0.421 accuracy/F1 at the subcategory level, although its binary F1 remained lower than that of BERTić.

Since the held-out evaluation provides only a single test estimate for the supervised BERTić baseline, we additionally performed stratified 5-fold cross-validation on the development portion of the dataset. The held-out test subset was excluded from this procedure and retained for direct comparison with the LLM-based configurations. The cross-validation results are reported as mean ± standard deviation across folds in Table XVI.

TABLE XVI
STRATIFIED 5-FOLD CROSS-VALIDATION RESULTS FOR BERTIĆ

Task	Accuracy	F1
Binary detection	0.737 ± 0.022	0.649 ± 0.020
Category classification	0.702 ± 0.033	0.456 ± 0.026
Subcategory classification	0.641 ± 0.026	0.375 ± 0.037

The stratified 5-fold cross-validation results provide additional evidence regarding the stability of the supervised BERTić baseline. The binary detection task achieved the most stable performance, with an average F1-score of 0.649 ± 0.020. Category and subcategory classification remained more difficult, with average F1-scores of 0.456 ± 0.026 and 0.375 ± 0.037, respectively. These results further indicate that the supervised baseline is more stable for binary detection than for fine-grained label assignment, while the fixed held-out test subset remains the basis for direct comparison with the prompt-based LLM configurations reported in Table XIV.

The most important finding is the comparison with the original two-stage zero-shot approach. For Qwen3, few-shot prompting substantially improved the categorization stage. Category-level performance increased from 0.556 accuracy / 0.556 F1 in the zero-shot setting to 0.909 accuracy / 0.727 F1, while subcategory-level performance improved from 0.521 accuracy / 0.507 F1 to 0.818 accuracy / 0.714 F1. This indicates that Qwen3 benefited strongly from the addition of labeled demonstrations during fine-grained label assignment. This category-level improvement is also supported by McNemar's exact test, which showed a statistically significant difference between few-shot and zero-shot Qwen3 at the category level.

In contrast, Llama 3 showed a more moderate improvement. Relative to its zero-shot two-stage results, category-level performance increased from 0.333 accuracy / 0.333 F1 to 0.571 accuracy / 0.462 F1. However, at the subcategory level, performance decreased from 0.429 accuracy / 0.429 F1 in the zero-shot setting to 0.385 accuracy / 0.286 F1 with few-shot prompting. Thus, the effect of few-shot guidance appears to be strongly model-dependent and more beneficial for top-level category assignment than for fine-grained subcategory prediction in the case of Llama 3. The confidence intervals further indicate that the few-shot improvement is substantially more consistent for Qwen3 than for Llama 3, particularly for fine-grained categorization. This also suggests that the additional few-shot examples helped Qwen3 exploit the target taxonomy more reliably, while Llama 3 remained more sensitive to fine-grained subcategory distinctions.

In light of the architectural differences summarized in Table V, the comparative results suggest that the observed performance depends on both the prompting strategy and the capabilities of the underlying model. The comparative results suggest that the strong performance of certain LLMs cannot be attributed solely either to the underlying model or to the prompt design, but rather to their interaction. The few-shot experiments show that changing only the categorization prompt can substantially affect fine-grained performance, as illustrated by the strong improvements observed for Qwen3 relative to its zero-shot two-stage setting, while the gains for Llama 3 remain notably smaller. This indicates that prompt design plays an important role, but that its effectiveness is strongly conditioned by the capabilities of the underlying model.

E. LIMITATIONS AND ERROR SPACE ANALYSIS

While the proposed framework demonstrates robust performance and represents a significant advancement for automated content moderation in underrepresented regional contexts, several inherent limitations and challenges within the error space must be acknowledged. First and foremost, it is essential to emphasize that our objective is not to claim an absolute, universally superior classification paradigm across all NLP domains; rather, this study introduces a highly specialized, unique, and state-of-the-art framework specifically tailored for detecting and performing a fine-grained, 14-category and subcategory classification of short hate speech texts in the Serbian language.

A primary constraint of this research stems from the relatively small size of the compiled dataset. Serbian is traditionally categorized as a low-resource language characterized by a severe scarcity of publicly available, annotated datasets and benchmark infrastructures. The core complexity lies in the collection and manual annotation of short textual units that explicitly contain hate speech. Due to the aggressive content moderation policies and automated purification mechanisms employed by major social media platforms and online news portals, a vast majority of hateful content is routinely and manually purged from the internet. Consequently, a completely raw, unfiltered data collection approach would yield an exceptionally low percentage of hate speech, rendering the dataset highly imbalanced and practically unusable for rigorous research. To mitigate this, our data collection pipeline required extensive and meticulous preprocessing and targeted filtering, which inherently limits the overall volume of the collected samples but ensures their high semantic density and relevance.

Furthermore, a distinct operational choice in our methodology was the exclusive reliance on open-source, freely available LLMs configured to run locally via the Ollama platform. While this approach guarantees complete data privacy and eliminates reliance on costly commercial APIs, it introduces specific behavioral limitations. Local open-source models process and output the requested hate speech examples and classifications directly for analytical purposes. Conversely, mainstream, globally managed commercial LLMs implement strict alignment and safety guardrails that systematically filter, reject, or block any input containing offensive discourse. Instead of providing analytical utility for academic research, these global models merely return generic safety warnings stating that the text contains inappropriate content, making them fundamentally non-viable for deep linguistic error space analysis.

Our evaluation paradigm is strictly localized, representing a Serbian-only evaluation framework. While this focus allows the models to capture the deep sociopolitical, cultural, and morphological nuances unique to the Serbian linguistic landscape, cross-lingual generalization was not explicitly tested within the scope of this study. This localized focus is further complicated by the ethical dimensions of data sourcing; since contemporary digital moderation increasingly forces toxic behavior into implicit or highly localized slang, gathering and verifying authentic instances of regional hate speech remains a continuous challenge. These factors combined define the operational boundaries of our current model, establishing a foundational baseline for future cross-lingual and scaled expansion.

To complement the quantitative evaluation, we conducted an error-space analysis of the zero-shot two-stage results for LLaMA and Qwen. The goal of this analysis is not only to measure aggregate performance, but also to identify systematic failure patterns across binary detection, category and subcategory predictions, and sentence length. This is particularly important in hate speech detection, where false positives may lead to excessive moderation, while false negatives may allow harmful content to remain undetected.

At the binary level, the two models exhibit different error profiles. LLaMA is more prone to false positives, which account for 61% of its binary errors, meaning that it more frequently classifies non-hate sentences as hate speech. In contrast, Qwen shows a more conservative detection behavior, with 56% of its binary errors coming from missed hate speech. Among the sentences misclassified by both models, false positives account for 54% of shared errors, suggesting two recurring failure spaces: ambiguous or politically charged content that both models tend to over-classify as hate speech, and more implicit hate speech that both models fail to detect. Sentence length also affects performance, with longer sentences producing more errors for both models; this effect is especially visible for LLaMA, whose false-positive rate increases most strongly as sentence length grows.

At the category level, misclassifications are concentrated in a small number of categories rather than being uniformly distributed across the taxonomy. For LLaMA, socioeconomic and political hate speech is responsible for a large portion of category-level errors, and the model frequently assigns the racial/ethnic category to sentences whose true category is different. This suggests that LLaMA often treats the racial/ethnic label as a default category for hate-adjacent content when the distinction between categories is unclear. Qwen exhibits a different pattern: racial/ethnic hate speech accounts for the largest share of category-level errors, but this is mostly due to missed detections at the binary stage rather than incorrect category assignment after detection. Socioeconomic and political hate speech is also a major source of errors for Qwen, confirming that politically framed and class-related discourse remains difficult for both models. At the subcategory level, both models struggle with categories that do not have an alphabetic suffix in the taxonomy, such as religious, age-based, and sports-related hate speech. These cases are often incorrectly mapped to lettered subcategories, suggesting that categories without strong lexical prototypes are less visible to zero-shot reasoning and require either clearer prompting, more representative examples, or supervised adaptation.

These error patterns also raise important ethical considerations. False positives may lead to excessive moderation, unjustified content removal, or the suppression of legitimate criticism, satire, and political expression, while false negatives may allow harmful or discriminatory content to remain visible. In addition, the dataset contains a substantial share of politically framed and ethnicity-related discourse, which may cause models to over-associate certain lexical or contextual cues with hate speech and underperform on less frequent categories. Therefore, the presented system should be interpreted as a decision-support tool rather than a fully autonomous moderation mechanism, consistent with recent work emphasizing that content moderation decisions require contextual, cultural, and human-centered judgment [69]. In practical deployment, model predictions should be combined with human review, transparent appeal procedures, and periodic re-evaluation on newly collected data to reduce bias, monitor category-specific errors, and limit potential harms caused by misclassification.

VI. CONCLUSION

This paper presented a comprehensive study of hate speech detection and classification in Serbian, a low-resource language, with a particular focus on the applicability of LLMs under prompt-based inference. In addition to constructing and analyzing a carefully curated paragraph-level dataset annotated at sentence granularity, we conducted an extensive comparative evaluation of multiple open-source LLMs across different task formulations, including single-stage and two-stage sentence-level pipelines, as well as paragraph-level analysis. The novelty of the study lies in combining Serbian-specific dataset construction, fine-grained hate speech taxonomy, local open-source LLM inference, multiple prompting strategies, external baselines, and statistical validation within a single comparative framework.

The dataset introduced in this work captures realistic discourse patterns observed in online communication, where neutral, offensive, and hate speech frequently co-occur within the same paragraph. By adopting sentence-level annotations within paragraph-level samples, the dataset enables fine-grained analysis while preserving broader contextual information. The accompanying statistical analysis demonstrates substantial diversity in paragraph length, hate speech density, and category distribution, reflecting the complexity of real-world hate speech phenomena in social networks as well as Serbian online media.

While the single-stage sentence-level approach proved adequate for binary hate speech detection, it consistently struggled with reliable category and subcategory classification. Most models improve under the two-stage design, with some exceptions (e.g., LLaMA and Mistral at category-level). This confirms that decoupling binary detection from fine-grained categorization reduces task interference and allows models to focus on more precise semantic distinctions.

Paragraph-level evaluation further highlighted the importance of contextual information. Access to multi-sentence context significantly improved category-level classification performance, particularly for models with stronger long-context reasoning capabilities. However, fine-grained subcategory classification at the paragraph level remained challenging, as overlapping or heterogeneous signals within a paragraph can obscure sentence-specific distinctions.

Across all experimental settings, model behavior exhibited notable variability. Among individual models, Qwen and DeepSeek incurred the highest inference latency, while the ensemble was the slowest overall due to executing all models per sample. Stronger performance was not strictly tied to extended context capacity: LLaMA 3, despite its smaller 8K context, achieved the strongest two-stage sentence-level detection performance, whereas Phi, even with a 128K context, showed inconsistent behavior across tasks (e.g., weak detection F1 despite strong subcategory scores). Overall, the additional experiments indicate that the observed LLM performance is shaped by both model capability and prompting strategy, with few-shot prompting improving results only when the underlying model can effectively exploit the added demonstrations.

Additional comparative experiments further showed that the effectiveness of fine-grained hate speech classification depends not only on the model family, but also on the prompting strategy. While the supervised BERTić baseline remained competitive for binary hate speech detection, it proved less effective for category-level and subcategory-level prediction in the present setup. Few-shot prompting, when introduced only in the categorization stage of the two-stage pipeline, produced strongly model-dependent effects: Qwen3 achieved substantial improvements over its zero-shot counterpart, whereas Llama 3 showed more limited gains. These findings suggest that few-shot prompting can significantly enhance structured label assignment for selected local LLMs, but its benefits are not uniform across models.

The ensemble strategy improved robustness most clearly in the two-stage sentence-level pipeline, achieving the best category-level classification performance, while remaining competitive at the subcategory level. These results indicate that the effectiveness of each model depends strongly on the task formulation and output granularity, and that accuracy and latency trade-offs are primarily driven by model-specific inference behavior rather than context capacity alone. Nevertheless, the substantial computational overhead associated with ensemble inference limits its applicability in real-time deployment scenarios.

Overall, this study provides empirical evidence that LLMs, even without task-specific fine-tuning, can serve as effective tools for hate speech detection in low-resource languages when guided by carefully designed prompts and evaluation strategies. Future work will focus on reducing inference latency, exploring hybrid human-in-the-loop annotation strategies, and extending the dataset and methodology to additional low-resource languages and cross-lingual settings.

ETHICS STATEMENT

All examples of hate speech included in this study were generated exclusively for research and model-training purposes. Their use is intended to support the development of AI-based tool capable of identifying and mitigating hate speech in the Serbian language. The authors explicitly condemn hate speech in all forms and aim to promote awareness and education regarding its harmful effects, both within Serbia and internationally.

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